

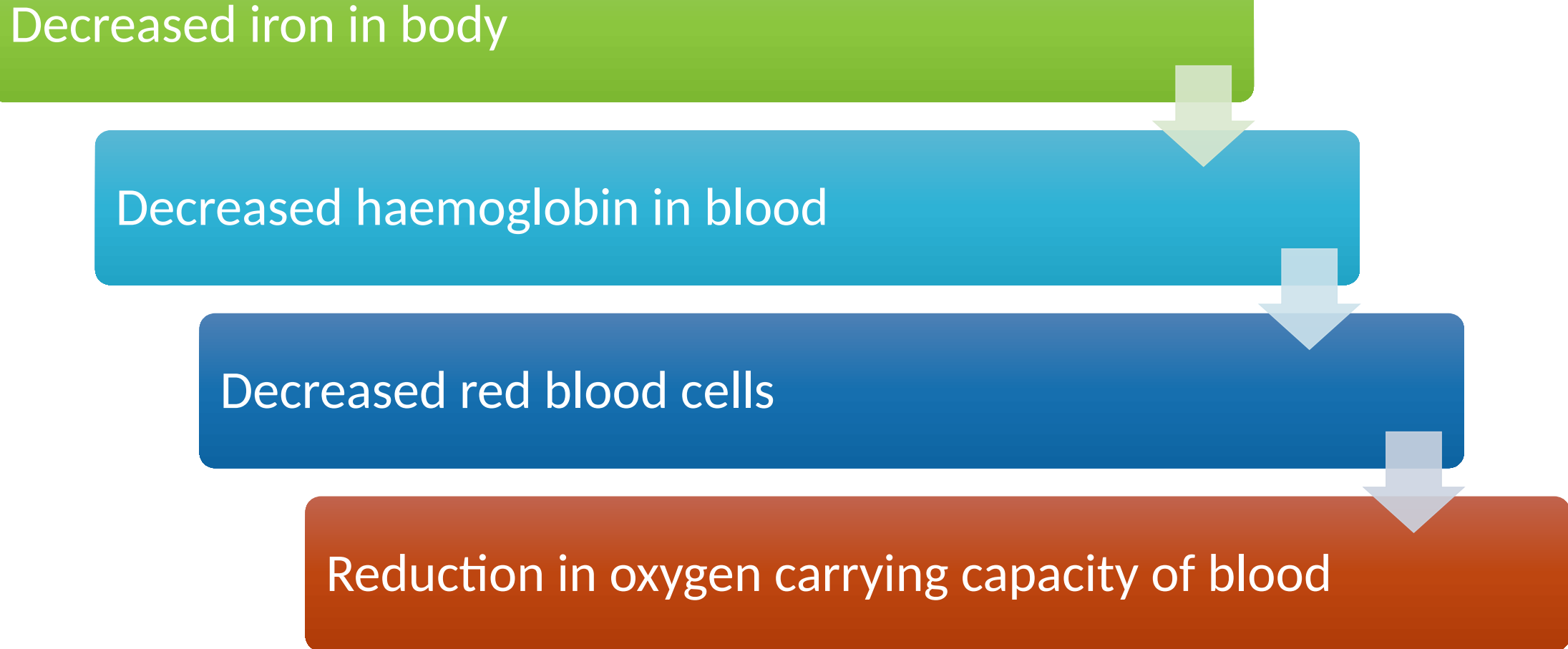
Iron deficiency Anemia in Pregnancy

Topic 2



What is Anaemia ?

Decreased iron in body



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graph TD; A[Decreased iron in body] --> B[Decreased haemoglobin in blood]; B --> C[Decreased red blood cells]; C --> D[Reduction in oxygen carrying capacity of blood];
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Decreased haemoglobin in blood

Decreased red blood cells

Reduction in oxygen carrying capacity of blood

Grades of anaemia in pregnancy

TYPE OF ANAEMIA	HAEMOGLOBIN CONCENTRATION
No Anaemia	≥ 11 gm/dl
Mild Anaemia	10 - 10.9 gm/dl
Moderate Anaemia	7 - 9.9 gm/dl
Severe Anaemia	< 7 gm/dl

Risk associated with Anemia in Pregnancy

Effect on mother

Abortions

Improper development of fetus

Increased rate of infections

Pre-eclampsia

Premature rupture of membranes

Preterm labour

Placental abruption

Postpartum haemorrhage

Puerperal sepsis

Effect on foetus and Newborn

Foetal Distress

Birth Asphyxia

Low birth weight

Prematurity

Symptoms of Severe Anaemia

Easy fatigue & weakness

Irregular Heartbeat

Shortness of breath

Dizziness or light headedness

Chest Pain

Cold hands and feet

Headache

Tingling sensation or leg cramps



Signs of severe Anaemia

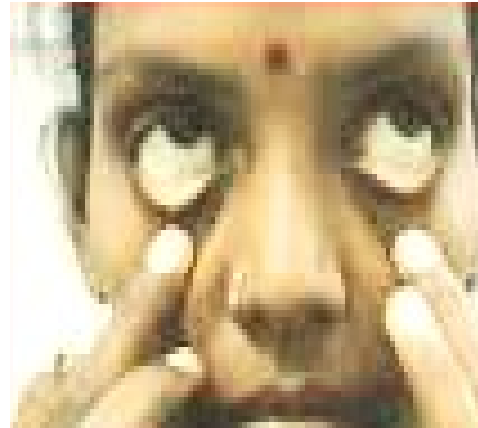
Pallor of conjunctiva, skin, tongue

Inflamed tongue (glossitis)

Inflamed mouth (stomatitis)

Brittle, pale nails

Generalized swelling



Diagnostic equipment for Anemia

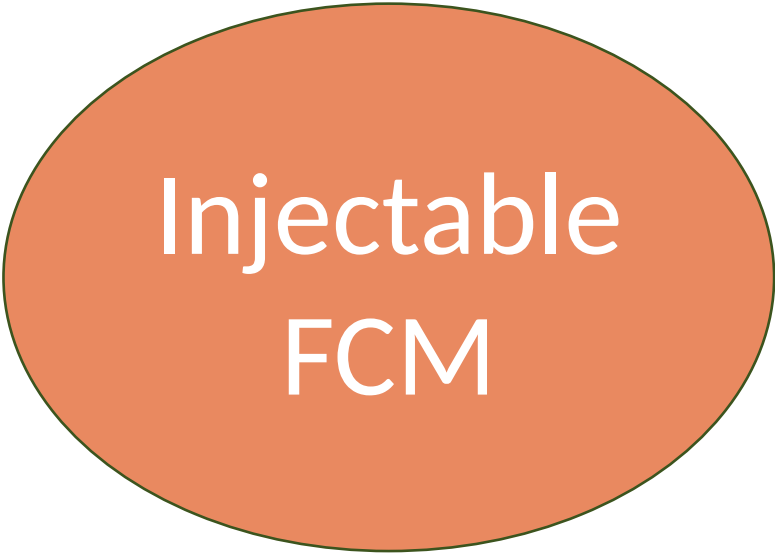
Level of Facility	Diagnostic equipment
VHSND	Hemo Cue
Facility Level ANC, PMSMA	CBC
Labour Room	Hemoglobinometer on Arrival, CBC as required

Methods of diagnosis

HemoCue

- The blood drop filled micro cuvette is placed in the cuvette holder and pushed to its measuring position
- After 15-60 seconds the hemoglobin value of the sample is displayed on the screen and should be recorded.





Injectable FCM

- Ferric Carboxy Maltose (FCM) is an intravenous iron formulation used to treat iron deficiency anemia (IDA), particularly in cases where
 - oral iron therapy has failed, is not tolerated,
 - or is unsuitable due to the severity of the anemia.
- FCM ensures effective iron delivery to bone marrow and other sites for hemoglobin synthesis.
- It replenishes iron stores quickly and raises hemoglobin levels

Indications for FCM

To be administered after 1st trimester for anaemia due to iron deficiency in the following conditions:

- Severe anaemia (Hb 5-6.9 g/dL) at 13-34 weeks of pregnancy
- Moderate anaemia (7 to 9.9 gm/dl) due to iron deficiency
 - Oral Iron not tolerated / Noncompliant
 - No improvement or improvement of less than 1 gm / dl after 1 month of oral IFA administration in IDA (Iron Deficiency Anaemia)
 - Moderate Aneamia after 32 weeks of gestation due to Iron deficiency aneamia.
- Can also be administered in lactating mothers if Hb 5-7.9 g/dL



Contraindications for FCM

- Iron overload Serum ferritin $>150 \mu\text{g/L}$
- Known hypersensitivity to iron preparation or excipients of Iron Sucrose or FCM
- Anemia not caused by iron deficiency
- Liver disorder like jaundice, cirrhosis, liver failure
- Acute cardiac failure
- Known case of thalassemia, sickle cell anemia, hemolytic anemia
- Fever, Acute infection/Illness



Comparison between Iron Sucrose and FCM ...1

Iron Sucrose

- Lower stability; relatively short half-life of 5-6 hours
- Rapid increase in hemoglobin levels, typically seen within one week of completion of the final dose of the treatment
- The expected increase in Hb level for PW with severe and moderate anemia is approximately 2.5 gm/dL and 1.6 gm/dL, respectively 3 weeks after complete calculated dose of IV-IS treatment
- Time for administration: 30 minutes

Ferric Carboxy Maltose

- Higher stability, longer half-life of about 16 hours
- Rapid replenishment of iron stores with significantly higher Hb rise typically seen within one week and sustained upto 3 to 6 weeks. FCM brings greater and faster Hb increase as compared to IS
- The expected increase in Hb level for PW with severe and moderate anemia is approximately 3.6 gm/dL and 2.74 gm/dL, respectively 4 weeks after complete calculated dose of FCM treatment
- Time for administration: 15 minutes

Comparison between Iron Sucrose and FCM ...2

Iron Sucrose

- Dose: Need of multiple visits to deliver the required dose
- Maximum dose in one sitting: 200 mg
- Maximum dose in one week: 600 mg
- Maximum dose in one pregnancy is 1000 mg i.e 5 doses of 200 mg Inj. Iron Sucrose over 2 weeks
- Administration: After preparation using 0.9% Normal Saline, the initial phase of the infusion should be at a rate of 20-30 drops per minute for the first 5 minutes, subsequently increasing to 80-90 drops per minute

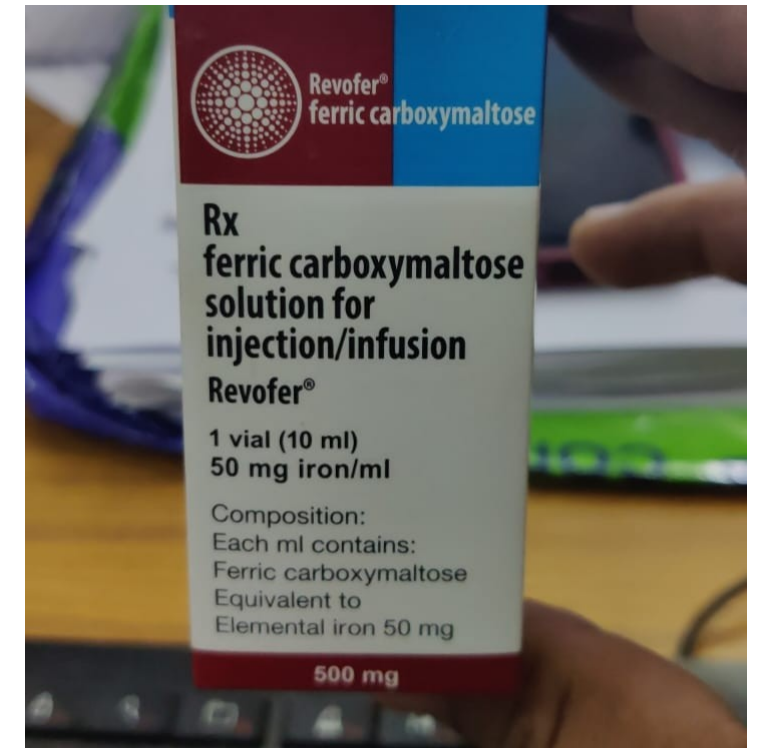
Ferric Carboxy Maltose

- Dose: Maximum dose and total replacement dose can be given in a single infusion,
- Maximum dose in one sitting: 1000 mg
- Maximum dose in one pregnancy is 1500 mg
- Administration: After preparation using 0.9% Normal Saline, the initial phase of the infusion should be of 20-30 drops per minute for the first five minutes and after that drop rate can be increased to 100-120 drops per minute

Dose Calculation of FCM

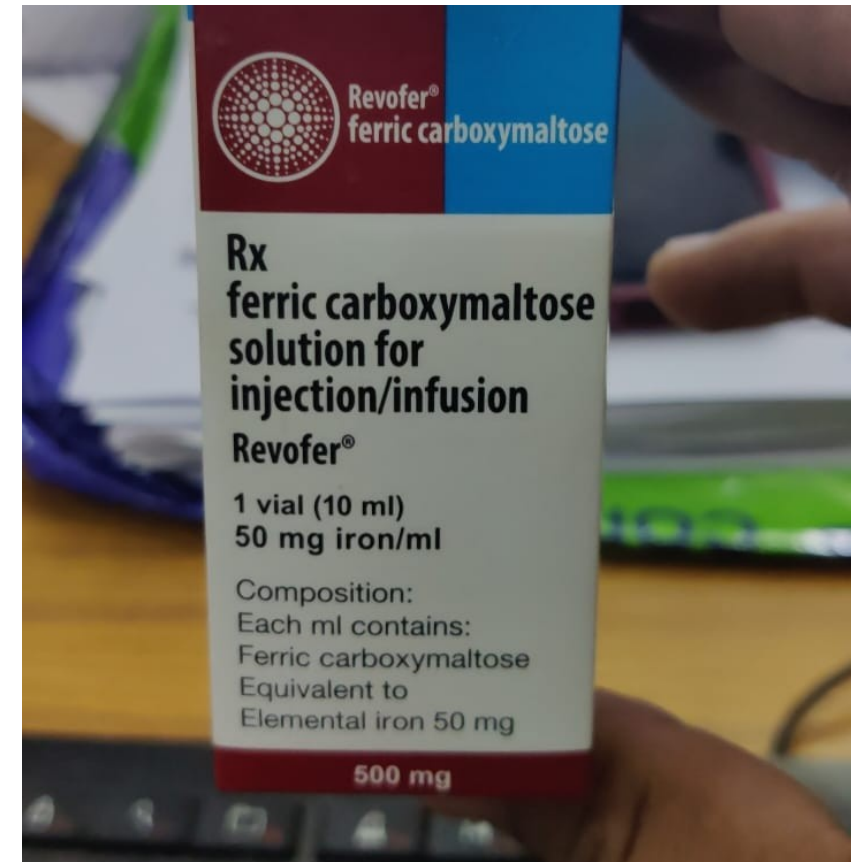
Total Iron deficit (mg)= 2.4 x Body wt (kg) x (11 – actual Hb in gm/dl) + 500

- If wt is more than 35 kg allowance for store is 500
- If wt is less than 35 kg allowance for store is 15 mg per kg
- FCM is available in 10 ml (50 mg iron) & 20 ml vials
- Administer with 100 ml of 0.9% normal saline over 15 mins
- Maximum dose per session 1000 mg, subsequent doses can be planned on 7th and 14th days



Administration of FCM

- Administered as IV infusion under supervision of MO at PHC/CHC/ SDH/DHH/MCH
- Vitals to be monitored before, during and after administration
- BP, heart rate, respiratory rate, temperature, fetal heart rate
- Patency of canula to be checked by flushing normal saline to avoid discoloration of skin due to extravasation of FCM
- PW to be observed for 30 mins for adverse events following infusion
- Test dose is not required



Precautions for FCM administration

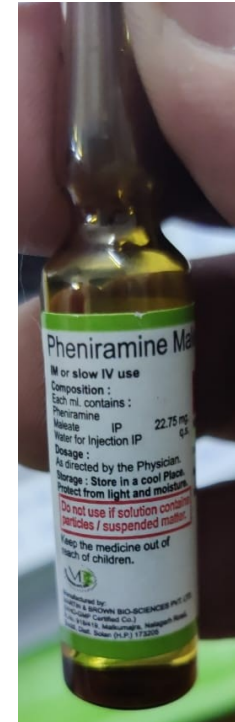
- Kit for managing adverse events must be kept ready
- Hand hygiene to be performed before and after procedure
- Sterile and disposable IV infusion set, IV canula and syringe must be used
- Patency of canula must be ensured to prevent extravasation of iron and permanent staining of skin.
- Remaining unused drug in the vial must be discarded

Management of adverse events following FCM administration

- Kit to be kept ready
 - 5 ml syringe – 1
 - 2 ml syringe – 1
 - Inj. Pheneramine Maleate – 1 ampoule
 - Inj. Hydrocortisone – 1 vial
 - Source of Oxygen

Mention batch number and expiry date of both drugs on the top of kit

- In case of adverse reaction, stop FCM administration, administer 1 amp (2ml) of Inj. Pheneramine Maleate IV, 1 vial of Inj. Hydrocortisone IV (100 mg in 5 ml).
- If not responding Inj. Adrenaline 0.5ml IM may be administered to relieve symptoms



FOLLOW UP ACTIONS AFTER ADMINISTRATION

- Check Hb level 4 weeks after administration
- If there is no improvement or less than 1 gm/dl increase in Hb, look for other causes of anemia
- IFA tablets need **not** be given for 3 months post full dose of infusion. Reevaluate after 3 months regarding course of action.
- Counsel the pregnant woman and family regarding appropriate diet.



Ferric Carboxymaltose Administration

Effectively Managing Iron Deficiency Anemia (IDA) in Pregnancy



Indications

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Process Flow



1. Haemoglobin testing



2. Doctors Visit



3. Inj. FCM Administration



4. Observation 30 Minutes



5. Haemoglobin check at 4 weeks of administration



iOS

Dose Calculation

Total Iron deficit (mg) = 2.4 x Body weight (kg) x (11 – actual Hb in gm/dl) + 500



Google Play

Points to remember

- FCM is available in 10ml (500 mg iron) & 20ml vials (1000mg).
- Administer with 100ml of 0.9% normal saline over 15 minutes.
- Maximum dose per session 1000mg, subsequent doses can be planned on 14th days.
- Maximum Dose per pregnancy 1500mg.

Developed by:

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